Active Directory RBAC & Least-Privilege Access Control

CONTROLLED LAB IMPLEMENTATION AND VALIDATION COMPLETE | October 2026

PROJECT OBJECTIVE

Implement Active Directory organizational structure, departmental identities, AGDLP group nesting, and restricted SMB/NTFS file-share permissions; verify authorized and unauthorized access using employee-context tests.

IMPLEMENTATION & VERIFIED RESULTS

Control / Activity	Verified result
Directory design	IAM-Lab OU with Finance, Human Resources, IT, and supporting OUs
User provisioning	6 fictional department accounts created
Role assignment	Finance, HR, and IT Global Security groups populated
AGDLP nesting	Global groups nested into matching Domain Local share groups
File server	IAM-APP01 departmental Finance, HR, and IT shares
NTFS and SMB controls	Department-specific Modify / Change rights; administrative and SYSTEM access retained
Access validation	9 of 9 expected outcomes matched across 3 representative users
Representative test users	ocarter: Finance only; sbennett: HR only; eparker: IT only

EVIDENCE & TECHNICAL DELIVERABLES

Documented OU hierarchy, membership and nesting screenshots, SMB and NTFS permission captures, and positive/negative access tests. Technical artifacts include an RBAC matrix, implementation guide, test results, Audit-IamShares.ps1, and Test-IamShareAccess.ps1.

LIMITATIONS & FOLLOW-UP

Nine observed outcomes cover three representative accounts, not all six provisioned accounts. An intermittent OPNsense default-route persistence issue was noted separately and remains outside the completed RBAC scope.

SKILLS DEMONSTRATED

Active Directory Domain Services, AGDLP, role-based access control, SMB and NTFS permissions, least privilege, PowerShell verification, and access-test evidence.

ASSESSMENT SCOPE

Results reflect a controlled training environment and the documented test cases; they do not represent a production deployment or formal security certification.